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Degrees:

Exp. 2027 Ph.D. in Physics – Astrophysics Track
Texas Christian University, Fort Worth, TX
May 2022 M.S. in Physics, California State University, Los Angeles, CA
(*Star Clusters on FIRE: Classifying Star Clusters Using Machine Learning*)
May 2020 B.A. in Physics (Minors: Math; Economics) Earlham College, Richmond, IN

Employment:

2025–present Research Assistant, Texas Christian University
2022–2025 Teaching Assistant, Texas Christian University
2021–2022 Teaching Assistant, California State University, Los Angeles

Awards:

2026 Best Graduate Student Poster - Student Research Symposium
Texas Christian University (\$200)
2021-2022 John Angione Endowed Scholarship for Physics,
California State University, Los Angeles (\$1,000)
2018-2020 Verne F. Swaim Memorial Scholarship
Earlham College (\$2,000 over 2 years)

Other Grants:

2025 SERC Grant – Radial and Azimuthal Galactic Chemical Gradients with the
OCCAM Survey and the Ages of Stellar Streams in SDSS-V
Texas Christian University (\$2,000)
2025 Graduate Student International Travel Grant
Texas Christian University (\$800)
2023-2024 Graduate Student Travel Grant
Texas Christian University (\$400)
2021-2022 Travel Support For Student Presentations
California State University, Los Angeles (\$300)

Observing Experience

2025–2026 McDonald Observatory, 2.7m/IGRINS 7 Nights Total (2 runs)
2024–2025 McDonald Observatory, 2.7m/Coudé 7 Nights Total (2 runs)

Teaching Experience

Teaching Assistant

2021-2025 PHYS 30123 Observational Astronomy Lab, TCU
PHYS 20484 General Physics II Lab, TCU (6 Sections)
PHYS 10154 General Physics I Lab, TCU (4 Sections)
PHYS 10293, 10263, 10273 Introductory Astronomy Labs, TCU (10 Sections)
PHYS 2200 Electricity, Magnetism and Circuits Lab, CSULA (2 Sections)

Publications:

First Author

1. **Otto, J.M.**, Frinchaboy, P.M., Donor, J., Medan, I., Myers, N., Hauck, A., Cunha, K., Zasowski, G., McKenzie, M., Zhao, J., Mèndez-Delgado, J.E., Sinha, A., Way, Z., Bizyaev, D., Casey, A.R., Fernández-Trincado, J.G., Pan, K. (2026), “*The Open Cluster Chemical Abundances and Mapping Survey XI. First Gradients from SDSS/MWM BOSS Determined Clusters*,” *Astronomical Journal*, *submitted*, doi: 10.48550/arXiv.2607.29023
2. **Otto, J.M.**, Frinchaboy, P.M., Myers, N.R., Johnson, J.W., Donor, J., Hossain, A., Mészáros, Sz., Wallace, H., Cunha, K., Bhattarai, B., Sinha, A., Zasowski, G., Loebman, S.R., Wiggins, A.I., Price-Whelan, A.M., Spoo, T., Souto, D., Bizyaev, D., Pan, K., Saydjari, A.K., Song, Y.-Y., (2025), “*The Open Cluster Chemical Abundances and Mapping Survey: VIII. Galactic Chemical Gradient and Azimuthal Analysis from SDSS/MWM DR19*,” *Astronomical Journal*, 171, 91, doi:10.3847/1538-3881/ae28d8.

Contributing Author

3. SDSS Collaboration, Aghakhanloo, M., Aguilar, D., **et al.**, (270 authors listed alphabetically, including **Otto, J.M.**), (2026), “*The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper*,” *Astrophysical Journal Supplement*, *submitted*, doi:10.48550/arXiv.2607.26149
4. Myers, N., Frinchaboy, P.M., Reggiani, H., Loebman, S., Manea, C., Shetrone, M., Cunha, K., Hawkins, K., Sinha, A., Zasowski, G., Donor, J., **Otto, J.M.**, Wiggins, A.I. (2026), “*The Open Cluster Chemical Abundances and Mapping Survey: X. Neutron Capture Abundances for Distant Open Clusters with Keck/HIRES and Magellan/MIKE*,” *Astronomical Journal*, *submitted*, doi:10.48550/arXiv.2607.00291
5. Medan I., Casey, A.R., Ji, A.P., **Otto, J.M.**, Sharifi, K., Way, Z., McKenzie, M., Myers, N.R., Stassun, K.G., Smith, P.J., Tkachenko, A., Chandra, V., Blanton, M.R., Frinchaboy, P.M., Stringfellow, G.S., Morrison, S. (2026), “*BOSS-CLAM: Utilizing a Constrained Linear Absorption Model to Infer Stellar Parameters from BOSS Spectra*,” *Astrophysical Journal*, *in press*, doi:10.48550/arXiv.2607.22822
6. Bijavara Seshashayana, S., Jönsson, H., D’Orazi, V., Cunha, K., Frinchaboy, P., **Otto, J.M.**, (2026), “*Age-dependent radial abundance gradients of 18 elements in Galactic open clusters from infrared MWM spectra*,” *Astronomy & Astrophysics*, *in press*, doi:10.48550/arXiv.2608.02563
7. Sinha, A., Zasowski, G., **Myers, N. R.**, Manea, C., Frinchaboy, P., et al., (2026), “*The Chemical Homogeneity of Single-Lined Spectroscopic Binaries in Open Clusters*,” *MNRAS*, 547, 440, doi:10.1093/mnras/stag440.
8. Nunnari, A., D’Orazi, V., Fiorentino, G., Braga, V.F., Bono, G., Fabrizio, M., Jönsson, H., Kudritzki, R.P., da Silva, R., Bergemann, M., Poggio, E., **Otto, J.M.**, Baeza-Villagra, K., Bragaglia, A., Ceci, G., Dall’Ora, M., Inno, L., Lardo, C., Matsunaga, N., Monelli, M., Sánchez-Benavente, M., Sneden, C., Tantaló, M., Thévenin, F., Kovtyukh, V., Di Criscienzo, M., Böcek Topcu, G. (2026), “*Classical Cepheids in the Galactic thin disk. I. Abundance gradients through NLTE spectral analysis*,” *Astronomy & Astrophysics*, 708, A17, doi:10.1051/0004-6361/202558288

9. SDSS Collaboration, Adamane Pallathadka, G., Aghakhanloo, M., Aird, J., Almeida, A., et al. (212 authors listed alphabetically, including **Otto, J.M.**), (2025), “*The Nineteenth Data Release of the Sloan Digital Sky Survey*,” *Astrophysical Journal Supplement*, 285, 9, doi:10.3847/1538-4365/ae4697.
10. Kollmeier, J. A., Rix, H.-W., Aerts, C., Aird, J., **et al.** (220 authors total listed alphabetically after first two authors, including **Otto, J.**), (2025), “*Sloan Digital Sky Survey-V: Pioneering Panoptic Spectroscopy*,” *Astronomical Journal*, 171, 52, doi:10.3847/1538-3881/ae0576.
11. Wiggins, A.I., Quinn, J.R., Oeur, M., Loebman, S.R., Frinchaboy, P.M., Daniel, K.J., McCluskey, F., **Otto, J.M.**, Woodward, H.R., D’Onghia, E., Wetzel, A., Parul, H., Bhattarai, B., Cozzi, M. (2025), “*Understanding the Origin and Dynamical Evolution of the Unique Open Star Cluster Berkeley 20 Using FIRE Simulations*,” *Astrophysical Journal Letters*, 995, L25, doi:10.3847/2041-8213/ae21bf
12. Spoo, T., Thomas, K., Toguchi-Tani, E.K., **Otto, J.**, Myers, N., et al., (2025), “*Extending the [C/N]-Age Calibration: Using Globular Clusters to Explore Older and Metal-Poor Populations*,” *Astronomical Journal*, 170, 333, doi:10.3847/1538-3881/ae14ec.
13. Mészáros Sz., Jofré, P., Johnson, J.A., Bird, J.C., Casey, A.R., Cunha, K., De Lee, N., Frinchaboy, P., Guiglion, G., Hegedű, V., Ji, A.P., Kollmeier, J.A., Ness, M.K., **Otto, J.**, Pinsonneault, M.H., Roman-Lopes, A., Sinha, A., Song, Y.-Y., Stringfellow, G.S., Stassun, K.G., Tayar, J., Tkachenko, A., Valentini, M., Way, Z., Weingrill, J. (2024), “*SDSS-V Milky Way Mapper (MWM): ASPCAP Stellar Parameters and Abundances in SDSS-V Data Release 19*,” *Astronomical Journal*, 170, 96 doi:10.3847/1538-3881/ade4b9

Contributed Talks:

- 2025 *The OCCAM survey: Radial and Azimuthal Gradients in the MW with Open Clusters*
Sloan Digital Sky Survey–V Collaboration Meeting - MPIA, Heidelberg, Germany
- 2024 *Reconstructing Star Formation Histories of Galactic Sub-structures using the [C/N]-Age Relationship*
Sloan Digital Sky Survey–V Collaboration Meeting - NMSU, Las Cruces, NM

Poster Presentations:

- 2026 *Understanding the Formation of the Milky Way’s Halo: Measuring the Star Formation History of Dwarf Galaxies and Tidal Streams*, AAS 247 Winter Meeting
- 2025 *Sifting through the Milky Way’s Star Clusters: Real or Impostor?*
SRS - Texas Christian University
- 2024 *Digging through the Galactic Graveyard: Chemistry and Ages of “Dead” Milky Way Satellite Galaxies*, AAS 243 Winter Meeting
- 2023 *Cluster Parameter Comparison between APOGEE and MWM/ASTRA using the Open Cluster Chemical Abundance and Mapping Survey*, RCA - Texas Christian University
- 2023 *The Open Cluster Chemical Abundance and Mapping (OCCAM) survey: Cross-calibration of APOGEE to MWM/ASTRA*
SDSS-V Collaboration Meeting - CCA, New York City, NY

- 2023 *Star Clusters on FIRE: Using Machine Learning to Classify Star Clusters in Synthetic Galaxy Images* - SRS - Texas Christian University
- 2022 *Star Clusters on FIRE: Classifying Star Clusters in Synthetic Galaxy Images Using Machine Learning*, AAS 240 Summer Meeting
- 2018 *Modeling Gravitational Waves in an Extreme Mass Ratio Inspiral System*, Epic Expo - Earlham College